

FOGLIO 10 TUTORATO

$$b) f(x) = x \sqrt{\frac{x-1}{x+1}}$$

$$\left\{ \begin{array}{l} \text{DOMINIO} \\ \sqrt{z} \quad z \geq 0 \end{array} \right.$$

$$\textcircled{1} \text{Dom}(f) = (-\infty, -1) \cup [1, +\infty)$$

$\uparrow$ $\lim$ $\uparrow$ $\lim$ $\uparrow$ $f(1)$ $\uparrow$ $\lim$

$$\left\{ \begin{array}{ll} \frac{x-1}{x+1} & x-1 > 0 \\ & x+1 > 0 \\ x+1 \neq 0 & x \neq -1 \end{array} \right.$$

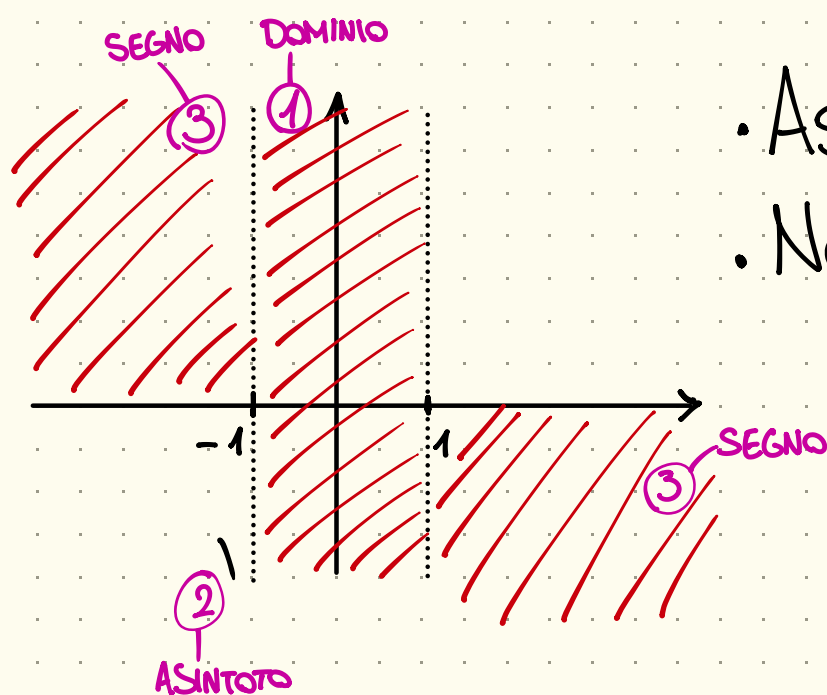
$$\lim_{x \rightarrow -\infty} \sqrt{\frac{x-1}{x+1}} =$$

$$= \lim_{x \rightarrow -\infty} x \cdot \sqrt{\frac{\cancel{x}(1 - \frac{1}{x})}{\cancel{x}(1 + \frac{1}{x})}} = -\infty$$

$\nearrow 0$ $\searrow 0$

$$f(1) = 1 \cdot \sqrt{\frac{1-1}{1+1}} = 0$$

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{x-1}{x+1}} = +\infty$$

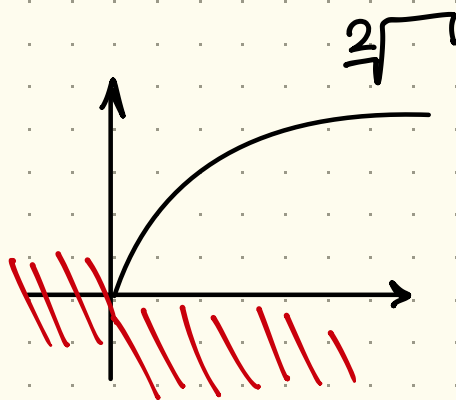


- $\textcircled{2}$
- ASINTOTO V. PER $x = -1$
 - NON CI SONO A.O.

SEGNO DI f (QUANDO $f(x) > 0$)

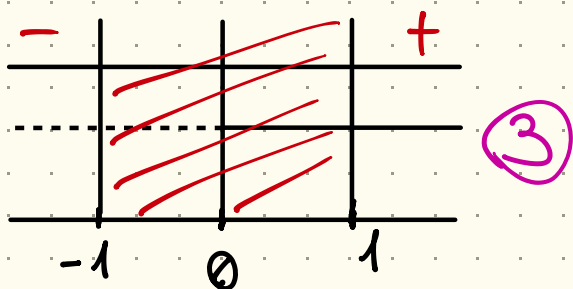
$$x > 0$$

$$\sqrt{\frac{x-1}{x+1}} > 0 \quad \forall x \in \text{Dom}(f)$$



$$\text{Dom}(f) = (-\infty, -1) \cup [1, +\infty)$$

É POSITIVA DA 1 A $+\infty$



DERIVATA DI f

$$f'(x) = \sqrt{\frac{x-1}{x+1}} + x \left(\sqrt{\frac{x-1}{x+1}} \right)'$$

$$\sqrt{\frac{x-1}{x+1}} = \left(\frac{x-1}{x+1} \right)^{\frac{1}{2}} \xrightarrow{\text{DERIVATO}} \frac{1}{2} \left(\frac{x-1}{x+1} \right)^{-\frac{1}{2}} \left[\frac{(x+1) - (x-1)}{(x+1)^2} \right]$$

$$z^{\frac{1}{2}}(x) \longrightarrow \frac{1}{2} z^{-\frac{1}{2}}(x) \cdot z'(x)$$

$$f'(x) = \left(\frac{x-1}{x+1} \right)^{-\frac{1}{2}} \left[\frac{x^2 + x + 1}{(x+1)^2} \right] > 0$$

$$\sqrt{\frac{x-1}{x+1}} > 0 \quad \begin{matrix} x^2 + x + 1 > 0 \\ (x+1)^2 > 0 \end{matrix}$$

